

1. Aşağıda verilen denklemlerin genel çözümlerini bulunuz.

- a) $(3D_x^2 - 13D_xD_y + 4D_y^2)z = 0$
 $[c : z = f_1(3y + x) + f_2(y + 4x)]$
- b) $(2D_x^2 + 7D_xD_y + 5D_y^2)z = 0$
 $[c : z = f_1(y - x) + f_2(2y - 5x)]$
- c) $(2D_x^2 - 3D_xD_y - 2D_y^2)z = 0$
 $[c : z = f_1(y + 2x) + f_2(2y - x)]$
- d) $(D_x^3 - D_xD_y^2)z = 0$
 $[c : z = f_1(y) + f_2(x + y) + f_3(y - x)]$
- e) $(3D_x^2D_y - 4D_xD_y^2)z = 0$
 $[c : z = f_1(x) + f_2(y) + f_3(3y + 4x)]$
- f) $(D_x^2D_y^2 + 2D_xD_y^3)z = 0$
 $[c : z = f_1(y) + f_2(x) + yf_3(x) + f_4(y - 2x)]$
- g) $(D_x^2D_y - 4D_xD_y^2 + 4D_y^3)z = 0$
 $[c : z = f_1(x) + f_2(y + 2x) + xf_3(y + 2x)]$
- h) $(4D_x^4 + 4D_x^3D_y + D_x^2D_y^2)z = 0$
 $[c : z = f_1(y) + xf_2(y) + f_3(2y - x) + xf_4(2y - x)]$
- i) $(7D_xD_y^3 - D_y^4)z = 0$
 $[c : z = f_1(x) + yf_2(x) + y^2f_3(x) + f_4(7y + x)]$
- j) $(5D_x^3D_y^2 + 6D_x^2D_y^3)z = 0$
 $[c : z = f_1(y) + xf_2(y) + f_3(x) + yf_4(x) + f_5(5y - 6x)]$
- k) $(D_x^3 - 6D_x^2D_y + 12D_xD_y^2 - 8D_y^3)z = 0$
 $[c : z = f_1(y + 2x) + xf_2(y + 2x) + x^2f_3(y + 2x)]$
- l) $(D_x^4D_y^2 + 3D_x^3D_y^3 + 3D_x^2D_y^4 + D_xD_y^5)z = 0$
 $[c : z = f_1(x) + yf_2(x) + f_3(y) + f_4(y - x) + xf_5(y - x) + x^2f_6(y - x)]$
- m) $(3D_x^2 + 2D_xD_y - 4D_y)z = 0$
 $[c : z = f_1(y) + e^{2y}f_2(3y - 2x)]$
- n) $(D_x^2 - 2D_xD_y + D_y^2 - D_x + D_y)z = 0$
 $[c : z = f_1(x + y) + e^x f_2(x + y)]$
- o) $(D_x^2 + 2D_xD_y + D_y^2 + 4D_x + 4D_y + 4)z = 0$
 $[c : z = e^{-2x} [f_1(y - x) + xf_2(y - x)]]$
- ö) $(D_x^3 + 2D_x^2D_y + D_xD_y^2 - D_x^2 - 2D_xD_y - D_y^2)z = 0$
 $[c : z = xf_1(y - x) + f_2(y - x) + e^x f_3(y)]$
- ö) $(2D_x^3 + D_x^2D_y - 12D_x^2 - 6D_xD_y + 18D_x + 9D_y)z = 0$
 $[c : z = f_1(2y - x) + e^{3x} [f_2(y) + xf_3(y)]]$

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- ö) $(D_x^3 + 2D_x^2D_y + D_xD_y^2 - D_x^2 - 2D_xD_y - D_y^2)z = 0$
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- ö) $(2D_x^3 + D_x^2D_y - 12D_x^2 - 6D_xD_y + 18D_x + 9D_y)z = 0$
 $[c : z = f_1(2y - x) + e^{3x} [f_2(y) + xf_3(y)]]$

- m) $(3D_x^2 - 2D_y^2 + D_x - 1)z = 3e^{x-y} \cos(x-y)$ $[c : z_p = e^{x-y} \sin(x-y)]$
n) $(D_x D_y^2 + D_x^2 - 2)z = 16e^{2x} \cos 3y$ $[c : z_p = -e^{2x} \cos 3y]$
o) $(D_x^2 + D_x D_y - 2D_y^2)z = 8 \sec^2(2x+y) \tan(2x+y)$ $[c : z_p = \tan(2x+y)]$
ö) $(D_x^2 + D_x D_y - D_x - D_y)z = \frac{1}{y^2}$ $[c : z_p = \frac{1}{y}]$
p) $(D_x^2 - D_y^2 - D_x + D_y)z = \frac{x+1}{x^2}$ $[c : z_p = -\log x]$
r) $(D_x^2 - D_y^2 + 4D_y - 4)z = 4x(1-y)$ $[c : z_p = xy]$
s) $(D_x^2 D_y - D_x D_y^2 + D_x D_y)z = 1$ $[c : z_p = xy]$

4. Aşağıdaki denklemlerin eliptik, parabolik ve hiperbolik oldukları böl-
geleri belirleyiniz.

- a) $z_{xx} - z_{xy} - 2z_{yy} = 0$ b) $z_{xx} + 2z_{xy} + z_{yy} = 0$
c) $2z_{xx} + 4z_{xy} + 3z_{yy} - 5z = 0$ d) $z_{xx} + 2xz_{xy} + z_{yy} + (\cos xy)z_x = z$
e) $yz_{xx} - 2z_{xy} + e^x z_{yy} + z = 3$ f) $e^{xy} z_{xx} + (\sinh x)z_{yy} + z = 0$
g) $xz_{xx} + 2xyz_{xy} - yz_{yy} - 2z = 0$ h) $xz_{xx} + 2xyz_{xy} + yz_{yy} = 0$

5. Aşağıdaki denklemlerin tiplerini belirleyerek karakteristiklerini bu-
lunuz ve kanonik forma indirgeyiniz.

- a) $z_{xx} - z_y + z = 0$ b) $3z_{xx} + 8z_{xy} + 4z_{yy} = 0$
c) $z_{xx} - z_{yy} + z_x + z_y = 0$ d) $z_{xx} + yz_{yy} = 0$
e) $z_{xx} - y^2 z_{yy} = 0$ f) $y^2 z_{xx} - 2xyz_{xy} + x^2 z_{yy} + yz_x + xz_y = 0$
g) $y^3 z_{xx} + z_{yy} = 0$ h) $yz_{xx} + z_{yy} = 0$

6. Aşağıdaki denklemleri kanonik forma indirgeyerek genel çözümlerini
bulunuz.

- a) $(n-1)^2 z_{xx} - y^{2n} z_{yy} = ny^{2n-1} z_y$
b) $y^2 z_{xx} - 2xyz_{xy} + x^2 z_{yy} = \frac{y^2}{x} z_x + \frac{x^2}{y} z_y$
c) $4z_{xx} - 8z_{xy} + 4z_{yy} = 1$
ç) $z_{xx} - 2z_{xy} + z_{yy} + 3(z_x - z_y) + 2z = (x+2y)^2$
d) $x^2 z_{xx} + 2xyz_{xy} + y^2 z_{yy} = 4x^2$
e) $z_{xx} - (2 \sin x)z_{xy} - (\cos^2 x)z_{yy} - (\cos x)z_y = 0$
f) $xy(z_{xx} - z_{yy}) + (x^2 - y^2)z_{xy} = yz_x - xz_y - 2(x^2 - y^2)$
g) $y^2 z_{xx} - 2yz_{xy} + z_{yy} = z_x + 6y$
h) $xz_{xx} - (x+y)z_{xy} + yz_{yy} - \frac{x+y}{x-y}(z_x - z_y) = 0$
ı) $x^2 z_{xx} + 2xyz_{xy} + y^2 z_{yy} = 5$
i) $4z_{xx} + 5z_{xy} + z_{yy} + z_x + z_y = 2$
j) $3z_{xx} + 10z_{xy} + 3z_{yy} = 3$

1. Aşağıda verilen başlangıç değer problemlerini çözünüz.

a) $U_{tt} - c^2 U_{xx} = 0$; $U(x, 0) = 0$, $U_t(x, 0) = 1$; $[c : U(x, t) = t]$

b) $U_{tt} - c^2 U_{xx} = 0$; $U(x, 0) = \sin x$, $U_t(x, 0) = x^2$;
 $[c : U(x, t) = \sin x \cos ct + x^2 t + \frac{1}{3} c^2 t^3]$

c) $U_{tt} - c^2 U_{xx} = 0$; $U(x, 0) = x^3$, $U_t(x, 0) = x$;
 $[c : U(x, t) = x^3 + 3c^2 x t^2 + x t]$

d) $U_{tt} - c^2 U_{xx} = 0$; $U(x, 0) = \cos x$, $U_t(x, 0) = e^{-1}$;
 $[c : U(x, t) = \cos x \cos ct + \frac{1}{e} t]$

e) $U_{tt} - c^2 U_{xx} = 0$; $U(x, 0) = \ln(1 + x^2)$, $U_t(x, 0) = 2$;
 $[c : U(x, t) = 2t + \frac{1}{2} (\ln(1 + x^2 + 2cxt + c^2 t^2) + \ln(1 + x^2 - 2cxt + c^2 t^2))]$

f) $U_{tt} - c^2 U_{xx} = 0$; $U(x, 0) = x$, $U_t(x, 0) = \sin x$;
 $[c : U(x, t) = x + \frac{1}{c} \sin x \sin ct]$

2. Aşağıda verilen homogen olmayan denklemlere ilişkin başlangıç değer problemlerini çözünüz.

a) $U_{tt} - U_{xx} = 1$; $U(x, 0) = \sin x$, $U_t(x, 0) = x$;
 $[c : U(x, t) = \frac{1}{2} [\sin(x + t) + \sin(x - t)] + xt + \frac{1}{2} t^2]$

b) $U_{tt} - c^2 U_{xx} = x + ct$; $U(x, 0) = x$, $U_t(x, 0) = \sin x$

c) $U_{tt} - c^2 U_{xx} = \sin x$; $U(x, 0) = \cos x$, $U_t(x, 0) = 1 + x$

d) $U_{tt} - c^2 U_{xx} = e^x$; $U(x, 0) = 5$, $U_t(x, 0) = x^2$;
 $[c : U(x, t) = 5 + x^2 t + \frac{1}{3} c^2 t^3 + \frac{1}{2c^2} (e^{x+ct} + e^{x-ct} - 2e^x)]$

e) $U_{tt} - c^2 U_{xx} = x e^t$; $U(x, 0) = \sin x$, $U_t(x, 0) = 0$;
 $[c : U(x, t) = \sin x \cos ct + (e^t - 1)(xt + x) - x t e^t]$

f) $U_{tt} - c^2 U_{xx} = 2$; $U(x, 0) = x^2$, $U_t(x, 0) = \cos x$;
 $[c : U(x, t) = x^2 + t^2(1 + c^2) + \frac{1}{c} \cos x \sin ct]$

3. İki ucu sabitleştirilmiş bir sicimin titreşim hareketini veren aşağıdaki başlangıç ve sınır değer problemlerini çözünüz.

a) $U_{tt}-c^2U_{xx}=0, 0 < x < 2, t > 0$ b) $U_{tt}-4U_{xx}=0, 0 < x < 1, t > 0$
 $U(x,0) = \sin\left(\frac{\pi x}{2}\right), 0 \leq x \leq 2$ $U(x,0) = 0, 0 \leq x \leq 1$
 $U_t(x,0) = 0, 0 \leq x \leq 2$ $U_t(x,0) = x(1-x), 0 \leq x \leq 1$
 $U(0,t) = U(2,t) = 0, t \geq 0$ $U(0,t) = U(1,t) = 0, t \geq 0$

c) $U_{tt}-3U_{xx}=0, 0 < x < 5, t > 0$ d) $U_{tt}-c^2U_{xx}=0, 0 < x < a, t > 0$
 $U(x,0) = 0, 0 \leq x \leq 5$ $U(x,0) = \cos x, 0 \leq x \leq a$
 $U_t(x,0) = (5-x)x, 0 \leq x \leq 5$ $U_t(x,0) = x+1, 0 \leq x \leq a$
 $U(0,t) = U(5,t) = 0, t \geq 0$ $U(0,t) = U(a,t) = 0, t \geq 0$

e) $U_{tt}-c^2U_{xx}=0, 0 < x < 1, t > 0$ f) $U_{tt}-c^2U_{xx}=0, 0 < x < \pi, t > 0$
 $U(x,0) = x(1-x), 0 \leq x \leq 1$ $U(x,0) = 3\sin x, 0 \leq x \leq \pi$
 $U_t(x,0) = 0$ $U_t(x,0) = 0$
 $U(0,t) = U(1,t) = 0, t \geq 0$ $U(0,t) = U(\pi,t) = 0, t \geq 0$

g) $U_{tt}-c^2U_{xx}=0, 0 < x < \pi, t > 0$ h) $U_{tt}-c^2U_{xx}=0, 0 < x < 1, t > 0$
 $U(x,0) = 0$ $U(x,0) = 0$
 $U_t(x,0) = 8\sin^2 x, 0 \leq x \leq \pi$ $U_t(x,0) = x\sin x, 0 \leq x \leq 1$
 $U(0,t) = U(\pi,t) = 0, t \geq 0$ $U(0,t) = U(1,t) = 0, t \geq 0$

ı) $U_{tt}-c^2U_{xx}=0, 0 < x < 1, t > 0$ i) $U_{tt}-c^2U_{xx}=0, 0 < x < \pi, t > 0$
 $U(x,0) = x(1-x), 0 \leq x \leq 1$ $U(x,0) = \sin x, 0 \leq x \leq \pi$
 $U_t(x,0) = x - \tan\frac{\pi x}{4}, 0 \leq x \leq 1$ $U_t(x,0) = x^2 - \pi x, 0 \leq x \leq \pi$
 $U(0,t) = U(1,t) = 0, t \geq 0$ $U(0,t) = U(\pi,t) = 0, t \geq 0$

j) $U_{tt}-c^2U_{xx}=0, 0 < x < 1, t > 0$
 $U(x,0) = U_t(x,0) = 0, 0 \leq x \leq 1$
 $U(0,t) = 3, t \geq 0$
 $U(1,t) = 5, t \geq 0$

4. $a \leq x \leq b$ aralığına yerleştirilen bir sicimin, sürtünmenin mevcut olduğu bir ortamdaki titreşimleri; bir dış kuvvetin bulunmadığı durumda

$$U_{tt} + 2\gamma U_t + \gamma^2 U - c^2 U_{xx} = 0 \quad (\gamma \text{ sabit})$$

denklemini ile verilir. Sicimin uç noktalarının sabitleştirildiğini ve titreşim hareketinin $U(x,0) = f(x), U_t(x,0) = g(x)$ başlangıç koşullarına uyduğunu varsayalım. $U(x,t) = e^{-\gamma t} w(x,t)$ dönüşümü yardımıyla sicimin hareketini belirleyen fonksiyonu bulunuz.